Lecture Notes

Lecture 14

Operators on Hilbert Space

This final lecture carries the operator theory of Part I—adjoints, the spectral theorem, the operator classes of Lecture 9—into the infinite-dimensional Hilbert spaces of Lecture 13, where it becomes the mathematical language of quantum mechanics. Two things change in infinite dimensions. First, eigenvalues no longer tell the whole story: an operator can be "non-invertible in the limit" without having any eigenvector, giving a *continuous spectrum*. Second, the operators of physics—position, momentum, energy—are *unbounded*, defined only on dense domains, and treating them rigorously is essential. We build operator norms and adjoints, classify the spectrum, meet position and momentum and their canonical commutation relation, and state the spectral theorem in the form quantum mechanics needs. The course closes with the dictionary between this linear algebra and the postulates of quantum theory.

Learning objectives

By the end of this lecture you will be able to:

- Use the operator norm and adjoint for bounded operators, carrying over the Lecture 9 classes.
- Classify the spectrum (point / continuous / residual) as the generalization of eigenvalues.
- Explain why position and momentum are unbounded and why $[\hat{x}, \hat{p}] = i\hbar$ has no finite model.
- State the spectral theorem (compact and general self-adjoint) and read off the QM dictionary.

1 Bounded operators and the adjoint

Definition 14.1 (Bounded operator; operator norm). A linear operator $T : \mathcal{H} \to \mathcal{H}$ is *bounded* if

$$\|T\| = \sup_{v \neq 0} \frac{\|Tv\|}{\|v\|} < \infty.$$

This is the operator norm of Lecture 10; what is new is that in infinite dimensions the supremum can be infinite, so boundedness is a genuine hypothesis. Bounded is equivalent to continuous, and the bounded operators form a Banach space $\mathcal{L}(\mathcal{H})$ under the operator norm.

Definition 14.2 (Adjoint). The *adjoint* $T^\dagger$ of a bounded operator is the unique bounded operator with $\langle T^\dagger u, v\rangle = \langle u, Tv\rangle$ for all u, v (it exists by the Riesz theorem). As in Lecture 9, T is *self-adjoint* if $T^\dagger = T$, *unitary* if $T^\dagger T = TT^\dagger = \mathrm{id}$, and *normal* if $T^\dagger T = TT^\dagger$.

These classes mean exactly what they did in finite dimensions, and the same proofs show self-adjoint operators have real expectation values and unitary operators preserve the inner product. What is new is everything to do with the spectrum.

Example 14.3 (Operator norms). The identity has $\|\mathrm{id}\| = 1$, and any nonzero *orthogonal* projection has norm 1 (an oblique idempotent can have norm larger than 1). A diagonal operator on ℓ^2 with entries d_n has norm $\sup_n |d_n|$, finite exactly when the d_n are bounded; multiplication by $m(x)$ on L^2 has norm $\sup|m|$. ◁

Example 14.4 (Bounded versus unbounded). Multiplication by $\sin x$ on $L^2(\mathbb{R})$ is bounded ($\|M\| = 1$); multiplication by x is not, since for ψ_n concentrated near $x = n$,

$$\frac{\|\hat{x}\psi_n\|}{\|\psi_n\|} \approx n \to \infty.$$

Boundedness is exactly a uniform bound on this ratio—which observables like position lack. ◁

Example 14.5 (Adjoints). The adjoint of multiplication by m is multiplication by $\overline{m}$, so a real-valued multiplier is self-adjoint. The right shift S on ℓ^2 has adjoint the left shift $S^\dagger(x_1, x_2, \dots) = (x_2, x_3, \dots)$, with $S^\dagger S = \mathrm{id}$ but $SS^\dagger \neq \mathrm{id}$—an isometry that is not unitary, impossible in finite dimensions. ◁

Example 14.6 (Self-adjoint, unitary, normal). Multiplication by a real function is self-adjoint; multiplication by $e^{i\theta(x)}$ is unitary; the Fourier transform is unitary. The Hamiltonian, position, and momentum are all self-adjoint—the operators whose spectra are physically observable values. ◁

Example 14.7 (A norm bound). For any bounded T, $|\langle u, Tv\rangle| \le \|u\|\,\|T\|\,\|v\|$ by Cauchy–Schwarz; one also has $\|T^\dagger\| = \|T\|$ and (stated without proof) $\|T^\dagger T\| = \|T\|^2$, the C^*-identity. These make $\mathcal{L}(\mathcal{H})$ a C^*-algebra, the abstract setting for quantum observables. ◁

2 The spectrum

Definition 14.8 (Spectrum). The *spectrum* of T is

$$\mathrm{spec}(T) = \{\lambda \in \mathbb{C} : T - \lambda\,\mathrm{id} \text{ is not invertible}\}.$$

A λ for which $T - \lambda\,\mathrm{id}$ has nontrivial kernel is an *eigenvalue* (the *point spectrum*); but $T - \lambda\,\mathrm{id}$ can fail to be invertible without having a kernel, giving the *continuous spectrum*.

In finite dimensions "not invertible" means "has a kernel," so the spectrum is exactly the eigenvalues. In infinite dimensions an operator can be injective with dense but not closed range—invertible on its range but not boundedly so—and such λ fill out a continuous spectrum with no eigenvector at all. For a self-adjoint operator the spectrum is real; for a unitary operator it lies on the unit circle—the infinite-dimensional shadows of Lecture 9.

Example 14.9 (Continuous spectrum). Multiplication by x on $L^2[0,1]$ has *no* eigenvectors: $x\psi(x) = \lambda\psi(x)$ forces $\psi = 0$ almost everywhere. Yet $\hat{x} - \lambda\,\mathrm{id}$ (multiplication by $x - \lambda$) is not boundedly invertible for any $\lambda \in [0,1]$, since $1/(x-\lambda)$ is unbounded there. So $\mathrm{spec}(\hat{x}) = [0,1]$, purely continuous spectrum. ◁

Example 14.10 (Pure point spectrum). A diagonal operator $De_n = d_n e_n$ on ℓ^2 has each d_n as an eigenvalue, so its point spectrum is $\{d_n\}$. The full spectrum is the closure $\overline{\{d_n\}}$: a limit point of eigenvalues belongs to the spectrum even when it is not itself an eigenvalue. ◁

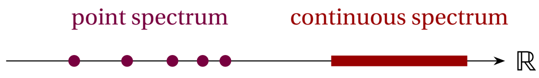

Figure 1: Two parts of a self-adjoint spectrum on the real line: isolated eigenvalues (the point spectrum, dots, possibly accumulating) and a continuous band (an interval, no eigenvectors). Both lie in $\operatorname{spec}(T)$.

Example 14.11 (Spectrum of the shift). The right shift on ℓ^2 has *no* eigenvalues, yet its spectrum is the entire closed unit disk $\{|\lambda| \le 1\}$ (a standard computation, stated here without proof). The gap between "eigenvalue" and "spectrum" is a purely infinite-dimensional effect. ◁

Remark 14.12 (The resolvent). The complement of the spectrum is the *resolvent set*, where $(T - \lambda\,\mathrm{id})^{-1}$ exists as a bounded operator. The resolvent is the analytic engine behind the spectral theorem and behind perturbation theory, where poles of $(H-\lambda)^{-1}$ locate energy levels. Concretely, with $R_0(z) = (H_0 - z\,\mathrm{id})^{-1}$, the Neumann series of Lecture 10 still applies and gives the *resolvent expansion*

$$\big(H_0 + \varepsilon V - z\,\mathrm{id}\big)^{-1} = \sum_{n\ge 0} (-\varepsilon)^n R_0(z)\big(V R_0(z)\big)^n,$$

valid whenever $|\varepsilon|\,\|V R_0(z)\| < 1$. Extracting the pole of this series at a level E_n reproduces the Rayleigh–Schrödinger formulas of Lecture 10.

Remark 14.13 (Perturbation theory in infinite dimensions). The *rigorous* half of Lecture 10 transfers verbatim; the *formal* half is what breaks. Spectral stability survives unchanged for bounded self-adjoint operators, because its only ingredient—that $\|(T - z\,\mathrm{id})^{-1}\| = 1/\operatorname{dist}(z, \operatorname{spec} T)$ for normal T—is a consequence of the spectral theorem, not of finite dimensionality. The Rayleigh–Schrödinger formulas likewise remain valid for an isolated eigenvalue of finite multiplicity, even when V is unbounded, provided V is H_0-bounded with relative bound less than one—the *Kato–Rellich* theorem, which covers the Coulomb potential. Three things genuinely change.

- *The sum becomes an integral.* The reduced resolvent $S_n = \sum_{m\ne n} P_m/(E_n - E_m)$ runs over the whole spectrum, so where a continuous part is present it is an integral $\int (E_n - \lambda)^{-1} dP(\lambda)$ against the projection-valued measure of Theorem 14.38.
- *The bound may be unavailable.* Most physical perturbations have $\|V\| = \infty$, so the disk estimate of Lecture 10 says nothing and relative boundedness must replace it.
- *An eigenvalue can dissolve.* An eigenvalue embedded in continuous spectrum may cease to be an eigenvalue for every $\varepsilon \ne 0$, becoming a *resonance* with a finite lifetime (Fermi's golden rule). The eigenvector does not merely move; it stops existing—with no finite-dimensional analogue.

Example 14.14 (A divergent series that works). The quartic anharmonic oscillator $H(\varepsilon) = \frac{1}{2}(\hat p^2 + \hat x^2) + \varepsilon \hat x^4$ has a Rayleigh–Schrödinger series with *zero* radius of convergence, yet its first few terms are numerically excellent. The reason is visible without any computation: for $\varepsilon < 0$ the quartic term drives the potential to $-\infty$, so $H(\varepsilon)$ is unbounded below and has no discrete spectrum at all. No disk about $\varepsilon = 0$ can be a domain of analyticity, however small. The series is *asymptotic* rather than convergent—the normal situation in quantum field theory, and the reason "perturbation theory works" and "the perturbation series converges" are different claims. ◁

Remark 14.15 (Residual spectrum—and why physicists never meet it). For completeness, the spectrum splits into three parts. If $T - \lambda\,\mathrm{id}$ has a kernel, λ is in the *point spectrum*; if it is injective with dense range but no bounded inverse, λ is in the *continuous spectrum*; and if it is injective with range that is not even dense, λ is in the *residual spectrum*. The right shift exhibits the third kind: $\lambda = 0$ is residual, since S is injective but its range misses e_1 by a whole dimension. For *normal* operators—in particular every self-adjoint observable and every unitary—the residual spectrum is empty: if the range of $T - \lambda\,\mathrm{id}$ were not dense, its orthogonal complement $\mathrm{null}(T^\dagger - \overline{\lambda}\,\mathrm{id})$ would be nonzero, and for a normal operator that forces $\mathrm{null}(T - \lambda\,\mathrm{id}) \neq \{0\}$ (Lecture 9), putting λ in the point spectrum instead. This is why the observables of quantum mechanics carry only point and continuous spectrum.

Example 14.16 (Norm equals spectral radius). For a bounded self-adjoint (or normal) operator, $\|T\| = \sup\{|\lambda| : \lambda \in \mathrm{spec}(T)\}$—proved in Lecture 10 for normal operators in finite dimensions, and still true here, though we quote rather than prove the general case. A self-adjoint operator with spectrum in $[m, M]$ thus has norm $\max(|m|, |M|)$—the spectrum controls the size, just as for Hermitian matrices. ◁

Example 14.17 (A computed spectrum). The operator $De_n = \frac{1}{n}e_n$ on ℓ^2 has eigenvalues $\{1/n : n \geq 1\}$ and spectrum $\{1/n\} \cup \{0\}$. The limit 0 lies in the spectrum—D is not invertible, since $De_n \to 0$—though it is not an eigenvalue. D is compact, with 0 its only spectral accumulation point. ◁

Example 14.18 (Spectrum of a multiplication operator). For multiplication by a continuous real g on $L^2[a, b]$, the spectrum is exactly the range (stated without proof),

$$\mathrm{spec}(M_g) = \overline{\{g(x) : x \in [a, b]\}},$$

self-adjoint because g is real. With $g(x) = x^2$ on $[-1, 1]$ this is $[0, 1]$, and there are no eigenvalues unless g is constant on a set of positive measure. ◁

3 Unbounded operators: position and momentum

The observables of quantum mechanics are typically unbounded. On $L^2(\mathbb{R})$ the *position* and *momentum* operators are

$$(\hat{x}\psi)(x) = x\,\psi(x), \qquad (\hat{p}\psi)(x) = -i\hbar\,\psi'(x),$$

each defined only on a dense domain (states for which the result is again in L^2). They are unbounded—no single constant C gives $\|\hat{x}\psi\| \leq C\|\psi\|$—yet self-adjoint on the right domains.

Theorem 14.19 (Canonical commutation relation). On a suitable dense domain,

$$[\hat{x}, \hat{p}] = \hat{x}\hat{p} - \hat{p}\hat{x} = i\hbar\ \mathrm{id}\,.$$

Example 14.20 (Deriving the commutator). Apply $[\hat{x}, \hat{p}]$ to a test function: $\hat{x}\hat{p}\psi = -i\hbar\,x\psi'$ and $\hat{p}\hat{x}\psi = -i\hbar(x\psi)' = -i\hbar(\psi + x\psi')$, so

$$[\hat{x}, \hat{p}]\psi = -i\hbar\,x\psi' + i\hbar(\psi + x\psi') = i\hbar\,\psi.$$

The $x\psi'$ terms cancel, leaving $i\hbar$ id. ◁

Example 14.21 (Momentum is diagonal in Fourier space). The plane wave $e^{ipx/\hbar}$ formally satisfies $\hat{p}\,e^{ipx/\hbar} = p\,e^{ipx/\hbar}$, but it is not in $L^2(\mathbb{R})$, so p is not an eigenvalue—it is continuous spectrum. The Fourier transform (Lecture 13) diagonalizes $\hat{p}$, turning $-i\hbar\,d/dx$ into multiplication by p; position and momentum are Fourier-conjugate multiplication operators. ◁

Example 14.22 (Ladder operators). In natural units set $a = \frac{1}{\sqrt{2}}(\hat{x} + i\hat{p})$. Then $[a, a^\dagger] = \mathrm{id}$ follows from $[\hat{x}, \hat{p}] = i\,\mathrm{id}$, and the number operator $N = a^\dagger a$ satisfies $[N, a^\dagger] = a^\dagger$, so $a^\dagger$ raises the energy by one quantum. The oscillator's whole spectrum is built algebraically from the commutator. ◁

Example 14.23 (Commutators build dynamics). From $[\hat{x}, \hat{p}] = i\hbar$ and the Leibniz rule $[A, BC] = [A, B]C + B[A, C]$,

$$[\hat{x}, \hat{p}^2] = 2i\hbar\,\hat{p}, \qquad [\hat{p}, \hat{x}^2] = -2i\hbar\,\hat{x}.$$

These give the Heisenberg equations $\dot{\hat{x}} = \hat{p}/m$ and $\dot{\hat{p}} = -V'(\hat{x})$ for a particle in a potential—Ehrenfest's theorem. ◁

Proposition 14.24 (No finite-dimensional model). There are no finite-dimensional matrices X, P with $[X, P] = i\hbar I$.

Proof. Taking traces, $\mathrm{tr}[X, P] = \mathrm{tr}(XP) - \mathrm{tr}(PX) = 0$ by cyclicity, while $\mathrm{tr}(i\hbar I) = i\hbar \dim \neq 0$. So the relation is impossible in finite dimensions: the canonical commutation relation forces an infinite-dimensional Hilbert space. ∎

Remark 14.25 (The uncertainty principle). The commutator $[\hat{x}, \hat{p}] = i\hbar\,\mathrm{id}$ feeds directly into the uncertainty bound of Lecture 9: $\Delta x\, \Delta p \geq \frac{1}{2}|\langle[\hat{x}, \hat{p}]\rangle| = \frac{\hbar}{2}$. Heisenberg's inequality is the Cauchy–Schwarz bound applied to two non-commuting observables.

Example 14.26 (Saturating the uncertainty bound). The oscillator ground state $\psi_0 = \pi^{-1/4}e^{-x^2/2}$ (natural units) has $\langle\hat{x}\rangle = \langle\hat{p}\rangle = 0$ and, by direct integration, $\Delta x^2 = \frac{1}{2} = \Delta p^2$, so

$$\Delta x\, \Delta p = \tfrac{1}{2},$$

which is $\hbar/2$ once units are restored. The Gaussian is the unique minimum-uncertainty state—the bound of Lecture 9 is attained. ◁

Remark 14.27 (Domains matter). For unbounded operators, $\langle Tu, v\rangle = \langle u, Tv\rangle$ on a domain makes T *symmetric*, but *self-adjoint* additionally requires the adjoint to share that domain. The distinction is not pedantic: only genuinely self-adjoint operators have a spectral theorem and generate unitary dynamics, so fixing the domain (and the boundary conditions) is part of defining an observable.

4 Compact operators and the discrete spectral theorem

The closest infinite-dimensional analogue of a Hermitian matrix is a *compact* self-adjoint operator.

Definition 14.28 (Compact operator). An operator $T \in \mathcal{L}(\mathcal{H})$ is *compact* if it is a norm-limit of finite-rank operators: there are operators T_N with finite-dimensional range and $\|T - T_N\| \to 0$. Equivalently, T maps the unit ball to a set in which every sequence has a convergent subsequence—T repairs, on its image, exactly the compactness failure of Lecture 13.

A diagonal operator $De_n = d_n e_n$ is compact iff $d_n \to 0$ (truncate to the first N entries: $\|D - D_N\| = \sup_{n>N}|d_n| \to 0$), and the identity is *not* compact—the unit ball itself is not compact, as the orthonormal sequence of Lecture 13 showed. Compactness is thus a strong form of "finite-dimensional up to small corrections," and it buys back the full finite-dimensional spectral theorem.

Theorem 14.29 (Spectral theorem, compact self-adjoint case · cited). A compact self-adjoint operator T on a Hilbert space has an orthonormal basis of eigenvectors $\{e_n\}$ with real eigenvalues $\lambda_n \to 0$, and

$$T = \sum_n \lambda_n |e_n\rangle\langle e_n|.$$

This is Lecture 10's spectral decomposition word for word, now with a countable sum of rank-one projections. Many physical Hamiltonians (a particle in a box, the harmonic oscillator through its resolvent) have exactly this structure: a discrete set of energy levels with an orthonormal basis of bound states.

Example 14.30 (An integral operator). On $L^2[0,1]$ consider the integral operator

$$(Tf)(x) = \int_0^1 k(x,y) f(y)\, dy$$

with a square-integrable kernel k; it is compact (Hilbert–Schmidt). If k is Hermitian, $\overline{k(y,x)} = k(x,y)$, then T is self-adjoint, so Theorem 14.29 applies: a real eigenvalue sequence $\lambda_n \to 0$ with an orthonormal eigenbasis. ◁

Example 14.31 (A spectral decomposition). A compact self-adjoint T with eigenpairs (λ_n, e_n) acts by

$$Tv = \sum_n \lambda_n \langle e_n, v\rangle\, e_n,$$

the infinite-dimensional version of $T = PDP^{-1}$ (Lecture 6). Truncating the sum gives finite-rank operators converging to T in norm—how compact operators are handled in practice. ◁

Example 14.32 (Diagonalizing a compact operator). For a compact self-adjoint T with $Te_n = \lambda_n e_n$ and $v = \sum_n c_n e_n$,

$$Tv = \sum_n \lambda_n c_n e_n, \qquad \|Tv\|^2 = \sum_n |\lambda_n|^2 |c_n|^2, \qquad \langle v, Tv\rangle = \sum_n \lambda_n |c_n|^2.$$

Every quantity reduces to a weighted sum over the eigenbasis—exactly a diagonal matrix, now with infinitely many entries. ◁

Example 14.33 (Eigenvalues going to zero). The integral operator on $L^2[0,1]$ defined by

$$(Tf)(x) = \int_0^1 \min(x,y)\, f(y)\, dy$$

is compact and self-adjoint, with eigenvalues $\lambda_n = \big((n - \tfrac{1}{2})\pi\big)^{-2} \to 0$. The decay of λ_n to 0 is the hallmark of compactness, and forces the spectrum to accumulate only there. ◁

Example 14.34 (The number operator). On ℓ^2 the number operator $N|n\rangle = n|n\rangle$ is self-adjoint with eigenvalues $0, 1, 2, \ldots$ and the number states as eigenbasis. It is neither bounded nor compact (its eigenvalues run to ∞); but its resolvent $(N+1)^{-1}$, with eigenvalues $1/(n+1) \to 0$, *is* compact—the usual route to a discrete spectrum. ◁

Example 14.35 (Resolvent of the number operator). The resolvent $(N+1)^{-1}$ acts on the number states by $|n\rangle \mapsto \frac{1}{n+1}|n\rangle$, with eigenvalues $1, \frac{1}{2}, \frac{1}{3}, \cdots \to 0$, so it is compact and self-adjoint with spectral decomposition $\sum_n \frac{1}{n+1}|n\rangle\langle n|$. Inverting a shifted unbounded operator is the standard way to expose a discrete spectrum. ◁

Example 14.36 (Particle in a box). For $-\frac{\hbar^2}{2m}\psi'' = E\psi$ on $[0, L]$ with $\psi(0) = \psi(L) = 0$, the solutions are $\psi_n(x) = \sqrt{2/L}\,\sin(n\pi x/L)$ with energies $E_n = \frac{\hbar^2\pi^2 n^2}{2mL^2}$. The ψ_n are the Fourier sine basis of $L^2[0, L]$ (Lecture 13)—a discrete spectrum with an eigenbasis, exactly the shape of Theorem 14.29. ◁

Example 14.37 (The harmonic oscillator). The oscillator Hamiltonian $H = \hbar\omega(N + \frac{1}{2})$ has spectrum $\{\hbar\omega(n + \frac{1}{2}) : n \geq 0\}$, a discrete ladder with eigenstates the number states $|n\rangle$. Built from the ladder operators $a, a^\dagger$, it is the canonical purely-discrete spectrum on an infinite-dimensional space. ◁

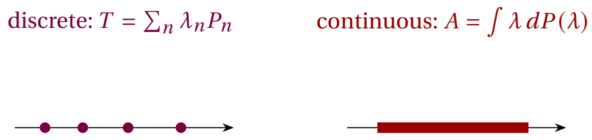

Figure 2: Two faces of the spectral theorem: a discrete-spectrum operator is a sum of projections onto eigenspaces; a continuous spectrum requires an integral against a projection-valued measure.

5 The spectral theorem and the postulates of quantum mechanics

For a general self-adjoint operator—bounded or unbounded, with continuous spectrum allowed—the sum becomes an integral.

Theorem 14.38 (Spectral theorem, general self-adjoint case · cited). Every self-adjoint operator A has a *projection-valued measure* P on its spectrum with

$$A = \int_{\mathrm{spec}(A)} \lambda \, dP(\lambda).$$

When the spectrum is discrete this is the sum of Theorem 14.29; a continuous spectrum gives a genuine integral.

Unitary dynamics follows: by Stone's theorem a self-adjoint Hamiltonian H generates the one-parameter unitary group $U(t) = e^{-iHt/\hbar}$, the solution of the Schrödinger equation. The whole apparatus assembles into the dictionary between linear algebra and quantum mechanics, the destination of this course.

Example 14.39 (Measurement and the Born rule). Writing $A = \int \lambda \, dP(\lambda)$, the projection-valued measure P assigns to each spectral region Ω the projection onto the matching states. Measuring A in state $|\psi\rangle$ gives a value in Ω with probability $\|P(\Omega)|\psi\rangle\|^2$, after which the state is $P(\Omega)|\psi\rangle$ (normalized)—the Born rule of Lecture 10, now valid for continuous spectra. ◁

Example 14.40 (Unitary time evolution). For $H|n\rangle = E_n|n\rangle$, Stone's theorem gives $U(t) = e^{-iHt/\hbar}$ acting by $|n\rangle \mapsto e^{-iE_n t/\hbar}|n\rangle$, so $\sum_n c_n|n\rangle$ evolves to $\sum_n c_n e^{-iE_n t/\hbar}|n\rangle$—the solution of the Schrödinger equation $i\hbar\,\partial_t\psi = H\psi$. ◁

Example 14.41 (A two-level system). For a spin in a field, $H = \frac{\hbar\omega}{2}\sigma_z$ has eigenstates $|0\rangle, |1\rangle$ with energies $\pm\frac{\hbar\omega}{2}$. The state $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ evolves to

$$\tfrac{1}{\sqrt{2}}\big(e^{-i\omega t/2}|0\rangle + e^{i\omega t/2}|1\rangle\big),$$

whose Bloch vector precesses about the z-axis at frequency ω—Larmor precession, the simplest unitary dynamics. ◁

Example 14.42 (Spectral decomposition of an observable). $\sigma_z = |0\rangle\langle 0| - |1\rangle\langle 1|$ is its own spectral decomposition: eigenvalues ± 1 with projections $|0\rangle\langle 0|$ and $|1\rangle\langle 1|$. Measuring σ_z projects onto $|0\rangle$ or $|1\rangle$, and a function acts as $f(\sigma_z) = f(1)|0\rangle\langle 0| + f(-1)|1\rangle\langle 1|$. The finite-dimensional spectral theorem is the seed of this whole chapter. ◃

Example 14.43 (A measurement probability). Measuring σ_z in $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ gives $+1$ with probability $|\langle 0, \psi\rangle|^2 = \frac{1}{2}$ and -1 with probability $\frac{1}{2}$, so $\langle \sigma_z \rangle = \frac{1}{2}(+1) + \frac{1}{2}(-1) = 0$—the average over the spectral projections. ◃

Example 14.44 (A one-parameter group). The evolution operators satisfy $U(s)U(t) = U(s+t)$ and $U(0) = \mathrm{id}$, each $U(t) = e^{-iHt/\hbar}$ unitary. Differentiating at $t = 0$ returns the generator, $i\hbar \frac{d}{dt} U(t)\big|_0 = H$, and Stone's theorem says every such group comes from a unique self-adjoint H. ◃

Example 14.45 (The free particle). The free Hamiltonian $H = \hat{p}^2/2m$ has *continuous* spectrum $[0, \infty)$ and no normalizable eigenstates—its "eigenfunctions" are the non-L^2 plane waves. Time evolution $e^{-iHt/\hbar}$ still makes sense through the spectral integral, spreading a wave packet. Discrete and continuous spectra coexist in one theory. ◃

Remark 14.46 (Two pictures). One may evolve states by $U(t)$ (the Schrödinger picture) or, equivalently, evolve observables by $A \mapsto U(t)^\dagger A U(t)$ (the Heisenberg picture). The two give identical predictions, related by a unitary—the same freedom as a time-dependent change of orthonormal basis.

Example 14.47 (Heisenberg equation of motion). In the Heisenberg picture an observable evolves by

$$\frac{d}{dt} A(t) = \frac{i}{\hbar} [H, A(t)],$$

the operator analogue of Hamilton's equations. For the free particle $[H, \hat{x}] = -i\hbar\, \hat{p}/m$ gives $\dot{\hat{x}} = \hat{p}/m$—the quantum statement that velocity is momentum over mass. ◃

Remark 14.48 (Functional calculus). The spectral theorem lets one form, for any reasonable function f,

$$f(A) = \int f(\lambda)\, dP(\lambda).$$

This defines the evolution $e^{-iHt/\hbar}$, the resolvent $(A - \lambda)^{-1}$, and projections onto energy bands. Functions of operators, introduced for matrices in Lecture 6, reach their final form here.

Remark 14.49 (Mixed states). The most general state is a *density operator*: a self-adjoint, positive, trace-one ρ, with $\langle A \rangle = \mathrm{tr}(\rho A)$. Pure states are the rank-one projections $|\psi\rangle\langle\psi|$; the reduced state of an entangled pair (Lecture 12) is a genuinely mixed ρ.

Example 14.50 (Expectation as a trace). For the qubit state $\rho = \frac{1}{2}(\mathrm{id} + \vec{r} \cdot \vec{\sigma})$ with Bloch vector $\vec{r}$,

$$\langle \sigma_z \rangle = \mathrm{tr}(\rho\, \sigma_z) = r_z,$$

so the Bloch coordinates are exactly the expectations of the Pauli observables. A pure state has $|\vec{r}| = 1$ (the Bloch sphere's surface); a mixed state lies strictly inside. ◃

Example 14.51 (Collapse). Measuring σ_z on $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ gives $+1$ with probability $|\alpha|^2$, after which

$$|\psi\rangle \longmapsto \frac{P_+|\psi\rangle}{\|P_+|\psi\rangle\|} = |0\rangle,$$

where $P_+ = |0\rangle\langle 0|$ is the spectral projection for eigenvalue $+1$. A repeat measurement now returns $+1$ with certainty—the projection postulate. ◃

Remark 14.52 (The dictionary). The course assembles into four correspondences: a *state* is a unit vector in a Hilbert space (up to phase); an *observable* is a self-adjoint operator; a *measurement* returns a spectral value with the Born-rule probability and projects the state; *dynamics* is the unitary group $e^{-iHt/\hbar}$. Every piece is linear algebra from the preceding lectures, set in an infinite-dimensional Hilbert space.

Remark 14.53 (Why infinite dimensions). The trace obstruction of Proposition 14.24 is no technicality: the canonical commutation relation, the continuous spectra of position and momentum, and the scattering states of the free particle all *require* an infinite-dimensional Hilbert space. Quantum mechanics is, irreducibly, the linear algebra of ℓ^2 and L^2.

Summary of main concepts

A bounded operator has finite operator norm and an adjoint defined through the Riesz theorem, and the self-adjoint, unitary, and normal classes of Lecture 9 persist unchanged. The spectrum $\operatorname{spec}(T) = \{\lambda : T - \lambda\,\mathrm{id} \text{ not invertible}\}$ generalizes the eigenvalues: in infinite dimensions it splits into a point spectrum (true eigenvalues) and a continuous spectrum (no eigenvector), with self-adjoint spectra real and unitary spectra on the unit circle. The observables of physics—position $\hat{x}$ and momentum $\hat{p} = -i\hbar\, d/dx$—are unbounded and obey $[\hat{x}, \hat{p}] = i\hbar\,\mathrm{id}$, which a trace argument shows is impossible in finite dimensions. Compact self-adjoint operators diagonalize exactly as Hermitian matrices do, $T = \sum_n \lambda_n |e_n\rangle\langle e_n|$; the general spectral theorem replaces the sum by an integral $\int \lambda\, dP(\lambda)$ against a projection-valued measure, and Stone's theorem turns a self-adjoint Hamiltonian into unitary time evolution $e^{-iHt/\hbar}$. This is the mathematics of quantum mechanics, and the end of our fourteen lectures.

- **Bounded operator**—finite operator norm = continuity; adjoint via Riesz; the self-adjoint, unitary, and normal classes persist.
- **Spectrum**—$\operatorname{spec}(T) = \{\lambda : T - \lambda\,\mathrm{id} \text{ not invertible}\}$; point (eigenvalues) versus continuous spectrum; self-adjoint real, unitary on the unit circle.
- **Unbounded observables**—$\hat{x}$ and $\hat{p} = -i\hbar\, d/dx$ obey $[\hat{x}, \hat{p}] = i\hbar\,\mathrm{id}$, impossible in finite dimensions (a trace argument).
- **Compact self-adjoint**—diagonalizes like a Hermitian matrix, $T = \sum_n \lambda_n |e_n\rangle\langle e_n|$ with $\lambda_n \to 0$.
- **General spectral theorem**—$A = \int \lambda\, dP(\lambda)$ against a projection-valued measure; Stone's theorem gives $e^{-iHt/\hbar}$.
- **QM dictionary**—states as unit rays, observables as self-adjoint operators, spectrum as outcomes, unitary time evolution.

Exercises for this lecture are in the companion sheet `exercises-14.pdf`.